

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
2.451	125.0	132.0	254.0	7199.154	28.343

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[65,2]	[59,5]	6.243	TIP	BRANCH
2	[7,16]	[59,5]	59.556	TIP	BRANCH
3	[59,5]	[61,28]	22.828	BRANCH	BRANCH
4	[139,7]	[138,11]	3.414	TIP	BRANCH
5	[148,18]	[138,11]	12.485	TIP	BRANCH
6	[131,16]	[138,11]	8.657	TIP	BRANCH
7	[400,11]	[370,49]	55.87	BRANCH	BRANCH
8	[400,11]	[438,15]	39.899	BRANCH	BRANCH
9	[400,11]	[414,33]	26.385	BRANCH	BRANCH
10	[438,15]	[443,17]	4.828	BRANCH	BRANCH
11	[438,15]	[414,33]	30.042	BRANCH	BRANCH
12	[443,17]	[462,17]	18.0	BRANCH	BRANCH
13	[443,17]	[454,64]	51.142	BRANCH	BRANCH
14	[462,17]	[496,36]	74.113	BRANCH	BRANCH
15	[462,17]	[479,35]	24.213	BRANCH	BRANCH
16	[38,22]	[39,25]	2.414	TIP	BRANCH
17	[39,25]	[61,28]	22.243	BRANCH	BRANCH
18	[21,29]	[39,25]	19.485	TIP	BRANCH
19	[61,28]	[66,45]	17.657	BRANCH	BRANCH
20	[414,33]	[414,54]	20.0	BRANCH	BRANCH
21	[293,52]	[300,35]	19.485	TIP	BRANCH
22	[259,39]	[300,35]	42.071	TIP	BRANCH
23	[321,24]	[300,35]	25.314	TIP	BRANCH
24	[479,35]	[496,36]	16.414	BRANCH	BRANCH
25	[479,35]	[454,64]	37.941	BRANCH	BRANCH
26	[496,36]	[469,75]	49.184	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[66,45]	[85,45]	18.0	BRANCH	BRANCH
28	[66,45]	[64,70]	24.828	BRANCH	BRANCH
29	[106,0]	[85,45]	55.042	TIP	BRANCH
30	[85,45]	[93,66]	22.899	BRANCH	BRANCH
31	[343,67]	[370,49]	36.385	TIP	BRANCH
32	[370,49]	[378,65]	18.314	BRANCH	BRANCH
33	[414,54]	[391,65]	26.556	BRANCH	BRANCH
34	[414,54]	[440,67]	29.971	BRANCH	BRANCH
35	[21,45]	[7,59]	24.314	TIP	BRANCH
36	[5,63]	[7,59]	3.828	TIP	BRANCH
37	[0,46]	[7,59]	14.485	TIP	BRANCH
38	[454,64]	[451,69]	5.243	BRANCH	BRANCH
39	[378,65]	[391,65]	12.0	BRANCH	BRANCH
40	[348,89]	[378,65]	39.698	TIP	BRANCH
41	[391,65]	[392,71]	5.414	BRANCH	BRANCH
42	[116,87]	[93,66]	38.213	TIP	BRANCH
43	[93,66]	[64,70]	29.657	BRANCH	BRANCH
44	[440,67]	[451,69]	10.828	BRANCH	BRANCH
45	[440,67]	[392,71]	55.698	BRANCH	BRANCH
46	[451,69]	[451,72]	2.0	BRANCH	BRANCH
47	[64,70]	[15,115]	70.154	BRANCH	BRANCH
48	[392,71]	[389,98]	27.243	BRANCH	BRANCH
49	[451,72]	[469,75]	18.243	BRANCH	BRANCH
50	[451,72]	[466,139]	74.456	BRANCH	BRANCH
51	[469,75]	[486,82]	18.485	BRANCH	BRANCH
52	[509,66]	[499,79]	16.142	TIP	BRANCH
53	[499,79]	[486,82]	13.243	BRANCH	BRANCH
54	[511,91]	[499,79]	16.556	TIP	BRANCH
55	[486,82]	[489,119]	37.414	BRANCH	BRANCH
56	[389,98]	[360,118]	36.284	BRANCH	BRANCH
57	[389,98]	[433,134]	58.083	BRANCH	BRANCH
58	[9,115]	[15,115]	5.0	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[1,81]	[9,115]	36.728	TIP	BRANCH
60	[3,123]	[9,115]	9.071	TIP	BRANCH
61	[15,115]	[28,133]	22.385	BRANCH	BRANCH
62	[360,118]	[322,127]	40.728	BRANCH	BRANCH
63	[360,118]	[360,140]	21.414	BRANCH	BRANCH
64	[501,116]	[489,119]	12.243	TIP	BRANCH
65	[489,119]	[472,139]	26.042	BRANCH	BRANCH
66	[302,115]	[322,127]	25.142	TIP	BRANCH
67	[322,127]	[318,151]	24.243	BRANCH	BRANCH
68	[28,133]	[51,141]	25.899	BRANCH	BRANCH
69	[15,209]	[28,133]	86.598	TIP	BRANCH
70	[433,134]	[455,143]	24.728	BRANCH	BRANCH
71	[433,134]	[400,154]	39.87	BRANCH	BRANCH
72	[466,139]	[472,139]	5.0	BRANCH	BRANCH
73	[466,139]	[455,143]	11.243	BRANCH	BRANCH
74	[472,139]	[481,164]	27.728	BRANCH	BRANCH
75	[360,140]	[400,154]	47.042	BRANCH	BRANCH
76	[360,140]	[353,147]	9.657	BRANCH	BRANCH
77	[60,144]	[51,141]	9.243	TIP	BRANCH
78	[51,141]	[56,180]	43.971	BRANCH	BRANCH
79	[455,143]	[435,185]	48.87	BRANCH	BRANCH
80	[353,147]	[351,171]	23.828	BRANCH	BRANCH
81	[353,147]	[318,151]	36.899	BRANCH	BRANCH
82	[318,151]	[315,152]	2.0	BRANCH	BRANCH
83	[257,128]	[315,152]	72.698	TIP	BRANCH
84	[315,152]	[310,212]	71.012	BRANCH	BRANCH
85	[400,154]	[398,169]	14.828	BRANCH	BRANCH
86	[481,164]	[491,182]	22.899	BRANCH	BRANCH
87	[481,164]	[435,185]	53.284	BRANCH	BRANCH
88	[398,169]	[380,172]	18.243	BRANCH	BRANCH
89	[398,169]	[432,185]	39.213	BRANCH	BRANCH
90	[351,171]	[329,195]	35.042	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[351,171]	[355,173]	3.414	BRANCH	BRANCH
92	[380,172]	[355,173]	27.728	BRANCH	BRANCH
93	[380,172]	[379,193]	20.414	BRANCH	BRANCH
94	[355,173]	[366,194]	24.556	BRANCH	BRANCH
95	[74,183]	[56,180]	18.243	TIP	BRANCH
96	[54,186]	[56,180]	5.828	TIP	BRANCH
97	[491,182]	[462,237]	80.941	BRANCH	BRANCH
98	[472,198]	[491,182]	24.627	TIP	BRANCH
99	[432,185]	[435,185]	2.0	BRANCH	BRANCH
100	[432,185]	[432,195]	9.0	BRANCH	BRANCH
101	[379,193]	[366,194]	12.414	BRANCH	BRANCH
102	[379,193]	[399,211]	29.385	BRANCH	BRANCH
103	[366,194]	[365,197]	2.0	BRANCH	BRANCH
104	[329,195]	[325,211]	16.657	BRANCH	BRANCH
105	[314,187]	[329,195]	19.971	TIP	BRANCH
106	[432,195]	[401,211]	36.627	BRANCH	BRANCH
107	[432,195]	[443,229]	37.142	BRANCH	BRANCH
108	[365,197]	[377,229]	36.556	BRANCH	BRANCH
109	[365,197]	[343,227]	38.698	BRANCH	BRANCH
110	[325,211]	[310,212]	14.414	BRANCH	BRANCH
111	[325,211]	[343,227]	23.213	BRANCH	BRANCH
112	[399,211]	[401,211]	1.0	BRANCH	BRANCH
113	[399,211]	[377,229]	29.213	BRANCH	BRANCH
114	[401,211]	[416,239]	32.799	BRANCH	BRANCH
115	[310,212]	[303,230]	19.899	BRANCH	BRANCH
116	[343,227]	[347,245]	18.657	BRANCH	BRANCH
117	[377,229]	[377,233]	3.0	BRANCH	BRANCH
118	[443,229]	[462,237]	21.314	BRANCH	BRANCH
119	[443,229]	[416,239]	29.728	BRANCH	BRANCH
120	[303,230]	[275,238]	30.314	BRANCH	BRANCH
121	[303,230]	[302,269]	41.728	BRANCH	BRANCH
122	[377,233]	[349,245]	31.971	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[378,240]	[377,233]	6.414	TIP	BRANCH
124	[462,237]	[448,273]	40.385	BRANCH	BRANCH
125	[275,238]	[266,264]	28.728	BRANCH	BRANCH
126	[222,222]	[275,238]	61.87	TIP	BRANCH
127	[416,239]	[404,277]	46.042	BRANCH	BRANCH
128	[347,245]	[349,245]	1.0	BRANCH	BRANCH
129	[347,245]	[329,273]	36.213	BRANCH	BRANCH
130	[347,245]	[323,270]	38.213	BRANCH	BRANCH
131	[349,245]	[357,297]	65.698	BRANCH	BRANCH
132	[225,270]	[223,259]	13.071	TIP	BRANCH
133	[222,257]	[223,259]	1.0	TIP	BRANCH
134	[225,258]	[223,259]	1.0	TIP	BRANCH
135	[242,275]	[266,264]	27.556	TIP	BRANCH
136	[266,264]	[275,275]	13.899	BRANCH	BRANCH
137	[302,269]	[308,270]	5.414	BRANCH	BRANCH
138	[302,269]	[275,275]	28.071	BRANCH	BRANCH
139	[308,270]	[323,270]	14.0	BRANCH	BRANCH
140	[300,289]	[308,270]	21.728	TIP	BRANCH
141	[323,270]	[329,273]	6.828	BRANCH	BRANCH
142	[329,273]	[329,284]	10.0	BRANCH	BRANCH
143	[448,273]	[480,282]	34.728	BRANCH	BRANCH
144	[448,273]	[423,288]	29.799	BRANCH	BRANCH
145	[275,275]	[270,305]	31.071	BRANCH	BRANCH
146	[404,277]	[413,288]	16.071	BRANCH	BRANCH
147	[404,277]	[391,310]	42.799	BRANCH	BRANCH
148	[480,282]	[500,282]	19.0	BRANCH	BRANCH
149	[480,282]	[470,329]	77.083	BRANCH	BRANCH
150	[510,261]	[500,282]	24.142	TIP	BRANCH
151	[506,289]	[500,282]	8.071	TIP	BRANCH
152	[329,284]	[350,297]	25.385	BRANCH	BRANCH
153	[329,284]	[317,306]	26.556	BRANCH	BRANCH
154	[413,288]	[423,288]	9.0	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[413,288]	[400,310]	25.971	BRANCH	BRANCH
156	[434,299]	[423,288]	15.314	TIP	BRANCH
157	[350,297]	[357,297]	6.0	BRANCH	BRANCH
158	[350,297]	[317,306]	42.042	BRANCH	BRANCH
159	[357,297]	[360,333]	37.899	BRANCH	BRANCH
160	[270,305]	[150,334]	132.012	BRANCH	BRANCH
161	[277,317]	[270,305]	13.485	TIP	BRANCH
162	[317,306]	[270,364]	80.569	BRANCH	BRANCH
163	[391,310]	[400,310]	8.0	BRANCH	BRANCH
164	[391,310]	[360,333]	39.698	BRANCH	BRANCH
165	[400,310]	[434,340]	49.698	BRANCH	BRANCH
166	[470,329]	[476,329]	5.0	BRANCH	BRANCH
167	[470,329]	[434,340]	39.142	BRANCH	BRANCH
168	[476,329]	[475,378]	55.042	BRANCH	BRANCH
169	[511,308]	[476,329]	42.87	TIP	BRANCH
170	[360,333]	[359,339]	5.414	BRANCH	BRANCH
171	[104,352]	[150,334]	55.627	TIP	BRANCH
172	[156,354]	[150,334]	22.314	TIP	BRANCH
173	[354,339]	[359,339]	4.0	BRANCH	BRANCH
174	[354,339]	[270,364]	94.598	BRANCH	BRANCH
175	[354,339]	[364,354]	31.971	BRANCH	BRANCH
176	[359,339]	[364,354]	15.657	BRANCH	BRANCH
177	[434,340]	[411,391]	60.113	BRANCH	BRANCH
178	[31,339]	[32,346]	6.414	TIP	BRANCH
179	[32,346]	[35,346]	2.0	BRANCH	BRANCH
180	[32,346]	[5,353]	28.485	BRANCH	BRANCH
181	[77,289]	[35,346]	76.497	TIP	BRANCH
182	[40,370]	[35,346]	25.485	TIP	BRANCH
183	[4,350]	[5,353]	2.414	TIP	BRANCH
184	[1,360]	[5,353]	7.657	TIP	BRANCH
185	[364,354]	[373,360]	10.485	BRANCH	BRANCH
186	[385,380]	[373,360]	23.971	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[397,344]	[373,360]	30.385	TIP	BRANCH
188	[251,379]	[270,364]	25.385	TIP	BRANCH
189	[193,376]	[220,379]	27.243	BRANCH	BRANCH
190	[191,373]	[193,376]	2.414	TIP	BRANCH
191	[184,385]	[193,376]	12.485	TIP	BRANCH
192	[475,378]	[460,397]	25.556	BRANCH	BRANCH
193	[475,378]	[487,380]	11.414	BRANCH	BRANCH
194	[220,379]	[275,435]	107.698	BRANCH	BRANCH
195	[237,362]	[220,379]	23.799	TIP	BRANCH
196	[509,417]	[487,380]	48.042	TIP	BRANCH
197	[509,347]	[487,380]	40.698	TIP	BRANCH
198	[411,391]	[408,401]	10.243	BRANCH	BRANCH
199	[411,391]	[392,398]	28.971	BRANCH	BRANCH
200	[297,389]	[305,395]	9.071	TIP	BRANCH
201	[359,385]	[305,395]	68.184	TIP	BRANCH
202	[303,399]	[305,395]	3.414	TIP	BRANCH
203	[460,397]	[434,417]	35.627	BRANCH	BRANCH
204	[483,418]	[460,397]	38.142	TIP	BRANCH
205	[392,398]	[408,401]	16.243	BRANCH	BRANCH
206	[392,398]	[379,422]	28.556	BRANCH	BRANCH
207	[408,401]	[408,415]	13.0	BRANCH	BRANCH
208	[408,415]	[434,417]	25.828	BRANCH	BRANCH
209	[408,415]	[379,422]	30.899	BRANCH	BRANCH
210	[494,481]	[434,417]	105.255	TIP	BRANCH
211	[379,422]	[378,424]	1.414	BRANCH	BRANCH
212	[378,424]	[337,429]	42.899	BRANCH	BRANCH
213	[449,488]	[378,424]	102.539	TIP	BRANCH
214	[331,426]	[331,428]	1.0	TIP	BRANCH
215	[331,428]	[337,429]	5.414	BRANCH	BRANCH
216	[331,428]	[311,432]	21.071	BRANCH	BRANCH
217	[337,458]	[337,429]	31.899	TIP	BRANCH
218	[311,432]	[275,435]	36.243	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[311,439]	[311,432]	6.0	TIP	BRANCH
220	[256,400]	[275,435]	42.456	TIP	BRANCH
221	[474,477]	[476,489]	11.828	TIP	BRANCH
222	[485,487]	[476,489]	8.828	TIP	BRANCH
223	[450,509]	[476,489]	35.213	TIP	BRANCH
224	[247,1]	[210,57]	80.113	TIP	TIP
225	[190,28]	[178,31]	13.243	TIP	TIP
226	[191,55]	[188,65]	11.243	TIP	TIP
227	[301,74]	[219,132]	110.711	TIP	TIP
228	[155,77]	[176,77]	21.828	TIP	TIP
229	[323,101]	[319,103]	4.828	TIP	TIP
230	[72,103]	[92,113]	24.142	TIP	TIP
231	[136,150]	[130,154]	7.657	TIP	TIP
232	[153,150]	[158,154]	6.657	TIP	TIP
233	[241,153]	[185,187]	71.255	TIP	TIP
234	[86,163]	[112,164]	31.971	TIP	TIP
235	[55,205]	[59,211]	7.657	TIP	TIP
236	[199,211]	[179,255]	52.284	TIP	TIP
237	[30,243]	[54,244]	26.071	TIP	TIP
238	[114,261]	[135,272]	65.113	TIP	TIP
239	[30,264]	[28,305]	44.314	TIP	TIP
240	[203,297]	[206,298]	3.414	TIP	TIP
241	[104,318]	[120,322]	18.485	TIP	TIP
242	[196,348]	[179,352]	18.657	TIP	TIP
243	[78,367]	[69,368]	9.414	TIP	TIP
244	[0,389]	[11,418]	33.556	TIP	TIP
245	[173,396]	[166,398]	7.828	TIP	TIP
246	[80,398]	[86,402]	7.657	TIP	TIP
247	[147,408]	[146,410]	2.414	TIP	TIP
248	[195,432]	[169,454]	36.284	TIP	TIP
249	[250,462]	[283,489]	56.042	TIP	TIP
250	[302,466]	[299,473]	8.243	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
251	[199,475]	[182,488]	23.556	TIP	TIP
252	[403,476]	[353,502]	81.012	TIP	TIP
253	[238,483]	[239,485]	2.414	TIP	TIP
254	[504,495]	[505,507]	13.243	TIP	TIP